D Y PATIL INTERNATIONAL UNIVERSITY
AKURDI PUNE

School of Computer Science, Engineering and Application(SCSEA)

B.Tech C.S.E 2nd year(2025-26)

Embedded System Design and Development

Project Report

Title:

AUTOMATIC PLANT WATERING SYSTEM

Name and PRN of Student group:

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1. Introduction

Need for the project:
People often forget to water their plants on time due to busy schedules. Irregular watering affects plant health and growth. An automatic watering system ensures plants receive water whenever the soil becomes dry, without human involvement.

Purpose:
The purpose of this project is to design an embedded system that senses soil moisture using a YL-38 module and automatically activates a 5V water pump when the soil is dry.

Usefulness:

- Ensures consistent plant watering
- Saves time and reduces manual effort
- Conserves water by watering only when needed
- Ideal for home gardens, offices, balconies, and educational projects

Reliability:
The system is powered directly through the Arduino (5V USB), which removes the need for external power supplies. Components like the YL-38 board, relay, and Arduino ensure stable and continuous performance.

Innovation:
This system replaces manual watering with real-time automation. LEDs and a buzzer provide instant feedback on soil status and pump activity, making it smarter and more interactive than simple irrigation systems.

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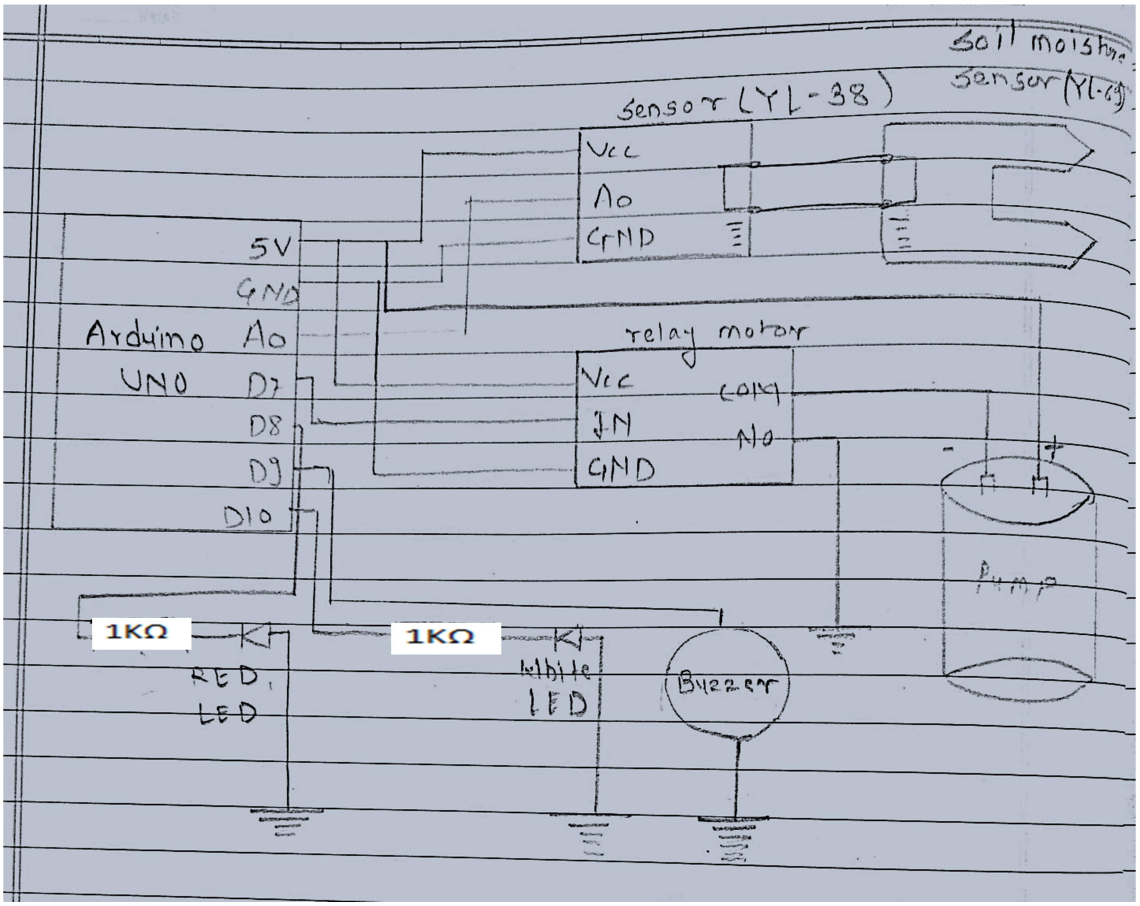
2. Circuit Diagram



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3. Hardware Components :

Component	Specification/Model	Qty	Purpose/Function
Arduino Uno	ATmega328P	1	Main microcontroller and power source
Soil Moisture Sensor	YL-38 (with YL-69 Probe)	1	Detects soil moisture level
Relay Module	5V Single Channel	1	Switches the pump ON/OFF
Mini Submersible Pump	5V DC	1	Pumps water to the plant
LED (Red, Green/White)	Standard	2	Indicators (soil dry, pump ON)
Resistors	1 kΩ	2	Current limiting for LEDs
Active Buzzer	5V Active	1	Audible alert when soil is dry
Jumper Wires	Male-to-Male	—	Circuit connections
Breadboard	—	1	Base for circuit wiring
Tubing Pipe	—	1	Water outlet from pump

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4. Working of the System

1. The YL-69 soil probe reads moisture level and sends analog output to the YL-38 board.
2. The YL-38 board outputs analog voltage to Arduino's **A0 pin**.
3. Arduino processes the moisture reading.
4. If soil is **dry**, Arduino:
    - Turns **ON the relay** to activate the pump
    - Turns **ON green LED**
    - Activates the **buzzer**
5. Once soil becomes **wet**, Arduino:
    - Turns **OFF the pump**
    - Turns **OFF the buzzer**
    - Turns **ON red LED** indicating soil is sufficiently moist

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5. Applications

- Smart home gardening
- Greenhouse environmental automation
- School/college embedded system projects
- Office plant maintenance
- Balcony/indoor plant irrigation

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6. Conclusion

The Automatic Plant Watering System is a low-cost, reliable, and fully automated solution for small-scale irrigation. By using a YL-38 moisture module, LEDs, buzzer, and relay-controlled water pump, the system ensures plants are watered efficiently without manual intervention. Its Arduino-based design makes it ideal for both practical use and educational demonstrations in embedded systems.